Boyang Li

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EDUCATION

New York University 2025 – 05/29
Ph.D. in Human-Centered Technology, Innovation & Design New York, NY

- Advisor: Takahiro Yabe
- Research Focus: Human-Centered AI, Causal Inference, Complex Systems, and Network Science

Peking University 2022 – 2025
M.Sc. in Geography, Smart Cities and Big Data Shenzhen, China

- Advisor: Wenjia Zhang
- Research Focus: Urban Computing, Inverse Reinforcement Learning, Spatial AI

Wuhan University 2018 – 2022
B.Eng. in Remote Sensing Science and Technology Wuhan, China

- Advisor: Zhipeng Gui
- Research Focus: Remote Sensing Change Detection, Visual SLAM, and UAV-based Geospatial Mapping

RESEARCH DIRECTIONS

- **Spatial Decision-Making & Brain-Inspired Spatial Intelligence**: cognitive maps, route choice, revealed preferences, inverse reinforcement learning, and topology-aware AI.
- **Urban Complex Systems & Agentic Collective Intelligence**: mobility, economic behavior, dependency networks, and emergent coordination in cities.
- **Representation Learning for Spatial AI**: n-dimensional positional encoding and geometry-aware embeddings for coordinates, images, videos, point clouds, and trajectories.
- **Spatial Reasoning & World Models**: geographic knowledge, spatial relations, and structured environment understanding in LLMs and AI agents.

EXPERIENCE

RUN Lab, New York University 2025 – Present
Ph.D. Researcher - Human-Centered AI for Complex Systems New York, USA

- Developing **LDP-DiffTraj**, diffusion-based generative models under local differential privacy for personalized human mobility trajectory synthesis.
- Proposing **nD-RoPE**, a generalized n-dimensional rotational positional encoding framework for Transformer architectures.
- Designing **EconAI**, an AI framework for modeling and simulating economic agents and strategic decision-making in complex systems.
- Developing **PDNets** (Predictive Dependency Networks) for modeling causal interactions and dynamic behavioral dependencies in complex urban systems.

BSAI Lab 2023 – 2025
Graduate Researcher - GeoAI & Human Environment Systems Shenzhen, China

- Proposed **MEDIRL-IC**, integrating individual heterogeneity into maximum-entropy inverse reinforcement learning for pedestrian route choice modeling.
- Developed cognitive map modeling frameworks (AVRIL) combining Bayesian IRL and topological data analysis to capture spatial decision-making structures.

- Designed **GeoRAG** and contributed to **AskNearby**, integrating geographic retrieval with LLMs for personalized neighborhood information and cognitive-map-aware recommendations.

Luojia-STC Lab Apr. 2021 – Jun. 2022
Undergraduate Researcher - Trajectory Prediction Wuhan, China

- Developed large-scale trajectory prediction models using deep inverse reinforcement learning
- Built WebGIS-based visualization system for spatial mobility analysis

PUBLICATIONS

** Equal contribution; † Corresponding author*

Li, Boyang, Yulin Wu, Sizhe Xu, Nuoxian Huang, Zhonghang Yuan, Shangyi Guo, Shu Yang, and Takahiro Yabe.

- *nD-RoPE: A Generalized RoPE for n-Dimensional Position Embedding*. In *Proceedings of the 43rd International Conference on Machine Learning (ICML 2026)*, PMLR 306. [Paper]

Li, Boyang, Shangyi Guo, Kota Tsubouchi, Toru Shimizu, Zhonghang Yuan, Joseph Y. J. Chow, and Takahiro Yabe.

- *LDP-DiffTraj: Diffusion Models under Local Differential Privacy for Personalized Human Mobility Trajectory Generation*. Accepted at the *34th ACM International Conference on Advances in Geographic Information Systems (SIGSPATIAL 2026)*. [Paper] [Best Paper Candidate]

Li, Boyang[*], Sizhe Xu[*], DongHak Lee, Shangyi Guo, Zhaonan Wang, and Takahiro Yabe.

- *Behaviorally Grounding Large Language Models with Revealed Choices for Human Mobility Prediction*. Accepted at the *34th ACM International Conference on Advances in Geographic Information Systems (SIGSPATIAL 2026)*. [Paper]

Li, Boyang, and Wenjia Zhang.

- *Enhancing Pedestrian Route Choice Models Through Maximum-Entropy Deep Inverse Reinforcement Learning With Individual Covariates (MEDIRL-IC)*. *IEEE Transactions on Intelligent Transportation Systems*, 25(12), 20446–20463. [Paper]

Li, Boyang, Yulin Wu, Nuoxian Huang, and Wenjia Zhang.

- *GridPE: A Grid Cell-Inspired Unified Position Embedding for Arbitrary-Dimensional Spaces*. *arXiv preprint arXiv:2406.07049*. [Paper]

Luyao Niu, Zhicheng Deng, **Boyang Li**[†], Nuoxian Huang, Ruiqi Liu, and Wenjia Zhang.

- *AskNearby: An LLM-Based Application for Neighborhood Information Retrieval and Personalized Cognitive-Map Recommendations*. In *Proceedings of the 8th ACM SIGSPATIAL International Workshop on AI for Geographic Knowledge Discovery (GeoAI 2025)*, 22–32. [Paper]

Wenjia Zhang, Daming Lu, Hongjin Liu, and **Boyang Li**.

- *Varying Built Environment Contexts and Trip Chain Decisions: A Multinomial-Choice Gradient Boosting Decision Trees Analysis*. *Travel Behaviour and Society*, 34, 100684. [Paper]

He Zhu, Wenjia Zhang, Nuoxian Huang, **Boyang Li**, et al.

- *PlanGPT: Enhancing Urban Planning with Tailored Language Model and Efficient Retrieval*. *arXiv preprint arXiv:2402.19273*. [Paper]

SELECTED CONFERENCE PRESENTATIONS

- **LDP-DiffTraj: Diffusion Models under Local Differential Privacy for Personalized Human Mobility Trajectory Generation**. Presentation, 34th ACM International Conference on Advances in Geographic Information Systems (SIGSPATIAL 2026), Riverside, CA, USA, 2026.
- **Distinct Regimes of Behavior-Based Dependency Networks Shape Urban Resilience**. Oral presentation, NetMob 2026, Niterói, Brazil, 2026.
- **nD-RoPE: A Generalized RoPE for n-Dimensional Position Embedding**. Poster presentation, 43rd International Conference on Machine Learning (ICML 2026), Seoul, South Korea, 2026.
- **AskNearby: An LLM-Based Application for Neighborhood Information Retrieval and Personalized Cognitive-Map Recommendations**. Oral presentation, ACM SIGSPATIAL GeoAI 2025 Workshop, Minneapolis, MN, USA, 2025.
- **Cognitive Anchors of Urban Residents: A Method Combining Topological and Euclidean Spatial Analysis**. Oral presentation, China Behavioral Geography Annual Conference, Shanghai, China, 2023.
- **Deep Inverse Reinforcement Learning-Based Travel Decision Analysis for Urban Residents**. Oral presentation, China Behavioral Geography Annual Conference, Xian, China, 2022.

GRANTS & FELLOWSHIPS

- NYU Tandon School of Engineering (SoE) Ph.D. Fellowship, fully funded (2025–2026)
- ACM SIGSPATIAL 2026 Student Travel Grant, $500 (2026)
- Yi-Innovator College Student Social Innovation Grant, RMB 120,000 (2024)
- Baidu Wenxin Cup Research Scholarship, RMB 15,000 (2024)
- Peking University Long Yuan Chang Ze Scholarship, RMB 5,000 (2024)
- Peking University Shenzhen Graduate School Tie Han Scholarship, RMB 8,000 (2023)
- Wuhan University First-Class Scholarship, RMB 3,000 (2019, 2021)
- Wuhan University Hong Tu Chuang Zhan Scholarship, RMB 5,000 (2020)